A Morse-Theoretic Construction of the Atiyah–Hirzebruch Spectral Sequence

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Contents

Notation & Conventions

This page collects all notational conventions used in this thesis. It is to be **removed before submission**.

1. Manifolds, Maps, and the Differential

Symbol	Meaning & Convention
M, N	smooth manifolds (always assumed without boundary unless stated)
T_pM, T_p^*M	tangent / cotangent space at $p \in M$
TM, T^*M	tangent / cotangent bundle
$\mathrm{d}f$	exterior differential of $f\colon M \to \mathbb{R}$; a 1-form on M
$\mathrm{d}_pf = (\mathrm{d}f)_p$	differential of f *at the point* p; a linear map $T_pM \to \mathbb{R}$. **Rule:** write d_pf (macro `\dat{p}f`) when evaluating at a specific point. Write $\mathrm{d}f$ for the full 1-form / bundle map.
d_pF	for $F\colon M \to N$: the linear map $T_pM \to T_{F(p)}N$. Also written $(\mathrm{d}F)_p$; never write $\frac{\mathrm{d}}{\mathrm{d}x}F$ for a map between manifolds.
$\frac{\mathrm{d}}{\mathrm{d}t}$	total derivative w.r.t. a *single* real variable t (roman d, macro `\diff`). Use for curves, flows evaluated on a trajectory.
$\frac{\partial}{\partial t}$	partial derivative; use only when *at least two independent* variables/parameters are present. **Never** write $\partial/\partial t$ for the time-derivative along a flow line.
$\nabla_v f$	covariant derivative of f in direction v (Levi-Civita or chosen connection)
$\mathrm{grad}_g f$	gradient of f with respect to Riemannian metric g; characterised by $g(\mathrm{grad}_g f, -) = \mathrm{d}f$

2. Flows and Dynamical Systems

Symbol	Meaning & Convention
$\varphi_t(x)$	flow of a vector field at time t through initial point x. **Convention:** always write φ_t, not $\varphi(t,\cdot)$; think of $\varphi_t \in \mathrm{Diff}(M)$ as a one-parameter family of diffeomorphisms.
$\varphi_t = \mathrm{e}^{tX}$	flow of the vector field X; $\varphi_0 = \mathrm{id}$.
$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x)$	time-derivative of the flow line $t \mapsto \varphi_t(x)$; equals $X(\varphi_t(x))$. Use roman d, never ∂.
$W^{\mathrm{u}}(p)$, $W^{\mathrm{s}}(p)$	unstable / stable manifold of a critical (rest) point p (macros `\Wu{p}`, `\Ws{p}`)

Symbol	Meaning & Convention
$\mathcal{M}(p,q)$	moduli space of (unparametrised) flow lines from p to q (macro `\Mflow{p}{q}`)

3. Restrictions and Evaluations

Core rule: distinguish carefully between a *fiber*, a *restriction to a subspace*, and a *value/evaluation.*

Symbol	Meaning & Convention
E_p	**fiber** of a vector bundle $E \to X$ over $p \in X$. **Preferred over** $E\|_{\{p\}}$ for fibers; use macro `E_p`.
$E\|_A$	**restriction** of the bundle E to the subspace $A \subseteq X$; this is itself a bundle over A. Macro: `\restr{E}{A}`.
$\xi\|_A$	section $\xi \in \Gamma(E)$ restricted to A; yields $\xi\|_A \in \Gamma(E\|_A)$.
$\xi(p)$ or ξ_p	**value** of the section ξ at the point p; an element of E_p. **Avoid** $\xi\|_p$ for this; use $\xi(p)$.
$\omega_p = \omega\|_p$	value of a differential k-form ω at p; an element of $\Lambda^k T_p^* M$. Acceptable to write $\omega\|_p$ only when the point-evaluation is being *emphasised* over the fiber.
Summary	E_p = fiber (bundle); $E\|_A$ = restricted bundle; $\xi(p)$ = value of section; ω_p = value of form.

4. Exterior Derivative, Coboundary, Connecting Homomorphism

Three distinct operators share similar notation; use them as follows.

Symbol	Macro	Use
∂	`\partial`	**Boundary operator** in chain complexes (homology direction). Default for singular chains; decorated for other theories (see §5).
d	`\diff`	**Exterior derivative** on differential forms: $\mathrm{d}\omega \in \Omega^{k+1}(M)$ for $\omega \in \Omega^k(M)$. Also the *total derivative* w.r.t. one real variable (e.g. $\mathrm{d}f/\mathrm{d}t$). **Never** use d as a boundary operator in a chain complex.

Symbol	Macro	Use
δ	`\delta`	**Two uses** (context distinguishes them): (1) *Coboundary operator* in a cochain complex (cohomology direction), e.g. in singular cohomology $\delta\colon C^k \to C^{k+1}$. (2) *Connecting homomorphism* in a long exact sequence of a pair or a short exact sequence of complexes, e.g. $\delta\colon H_n(X,A) \to H_{n-1}(A)$. Decorate when both appear simultaneously: δ^{conn} vs. δ^{cob}.

5. Chain Complexes and Homology Theories

Three homology theories appear simultaneously; they are always decorated.

Theory	Chain complex	Boundary / Homology
Singular	$(C_*(X), \partial)$	boundary ∂; homology $H_*(X)$. *Default / undecorated.*
CW	$(C_*^{\mathrm{CW}}(X), \partial^{\mathrm{CW}})$	boundary ∂^{CW}; homology $H_*^{\mathrm{CW}}(X)$. Macros: `\ChCW`, `\bCW`, `\HCW`.
Morse	$(C_*^{\mathrm{M}}(f), \partial^{\mathrm{M}})$	boundary ∂^{M}; homology $H_*^{\mathrm{M}}(f)$. Macros: `\ChM`, `\bM`, `\HM`.

Connecting homomorphisms in long exact sequences: always δ (decorated if needed: $\delta^{\mathrm{M}}, \delta^{\mathrm{CW}}$).

Filtration: $F^pC_*(X)$ for the filtration of the singular complex; boundary is still ∂.

6. Vector Bundles and K-Theory

Symbol	Meaning
$\mathrm{Vect}(X)$	set of isomorphism classes of complex vector bundles over X (macro `\Kvect{X}`)
$[E]$	stable isomorphism class / element in $\mathrm{K}(X)$
$\mathrm{K}(X) = \mathrm{KU}(X)$	complex K-theory (default); macro `\K`, `\KU`
$\mathrm{KO}(X)$	real K-theory; macro `\KO`
$\tilde{\mathrm{K}}(X)$	reduced complex K-theory
$E \oplus F,\ E \otimes F$	Whitney sum / tensor product of bundles
$E\vert_A$	restriction of bundle $E \to X$ to subspace A (same convention as §3)
E_p	fiber of E over p (same convention as §3)

7. Spectral Sequences

Symbol	Meaning
$E^r_{p,q}$	(p, q)-entry on the r-th page of a spectral sequence; macro `\Epq{r}{p,q}`
d^r	differential on the r-th page; $d^r\colon E^r_{p,q} \to E^r_{p-r,\,q+r-1}$ (cohomological convention); macro `\drpq{r}`
AHSS	Atiyah–Hirzebruch spectral sequence: $E^2_{p,q} \cong H^p(X; \mathrm{K}^q(\mathrm{pt})) \Rightarrow \mathrm{K}^{p+q}(X)$
$E^\infty_{p,q}$	abutment / E_∞-page

8. Frequently Used Macros (Quick Reference)

Macro	Output	Use
`\diff`	d	exterior deriv. / total deriv.
`\dat{p}`	d_p	deriv. at point p
`\restr{E}{A}`	$E\|_A$	restriction
`\Wu{p}`	$W^{\mathrm{u}}(p)$	unstable manifold
`\Ws{p}`	$W^{\mathrm{s}}(p)$	stable manifold
`\Mflow{p}{q}`	$\mathcal{M}(p, q)$	moduli of flow lines
`\ChM`	C^{M}_*	Morse chain complex
`\bM`	∂^{M}	Morse boundary
`\HM`	H^{M}_*	Morse homology
`\ChCW`	C^{CW}_*	CW chain complex
`\bCW`	∂^{CW}	CW boundary
`\HCW`	H^{CW}_*	CW homology
`\conn`	δ	connecting homom. (δ)
`\K`	K	K-theory
`\KO`	KO	real K-theory
`\KU`	KU	complex K-theory
`\Epq{r}{p,q}`	$E^r_{p,q}$	SS entry
`\drpq{r}`	d^r	SS differential
`\Crit`	Crit	critical point set
`\ind`	ind	Morse index
`\grad`	grad	gradient
`\euler`	e	Euler number (roman e)
`\ii`	i	imaginary unit (roman i)